

Setting up Intel INRC on LAVA

The following are frequently asked questions on how to install Intel INRC on LAVA. Please consult this file for questions on LAVA installation, transferring files to/from local machine and other issues.

How to login to Intel INRC cloud in Anaconda?

To log in to **Intel INRC Cloud** in **Anaconda**, follow these steps:

1. Sign in to Intel INRC Cloud

Go to the Intel Tiber AI Cloud and log in using your **INRC credentials**.

2. Set Up Anaconda for INRC Cloud

Once logged in, you need to configure Anaconda to work with Intel's cloud environment:

```
conda config --add channels https://software.repos.intel.com/python/conda
```

Then, install the required packages:

```
conda install intel-oneapi
```

3. Access Loihi Systems

If you're running **Lava** on **Loihi**, ensure you have access to **Intel's cloud-hosted Loihi systems**. You may need to request access through **INRC support**.

4. Run Lava on INRC Cloud

Once your environment is set up, you can run **Lava applications** on **Loihi**:

```
from lava.proc.lif.process import LIF
from lava.magma.runtime.run_configs import Loihi2HwCfg
from lava.magma.runtime.run_conditions import RunSteps

lif = LIF()
lif.run(RunSteps(num_steps=100), Loihi2HwCfg())
lif.stop()
```

If you encounter any issues, check the Intel INRC Cloud portal or contact **nrc_support@intel-research.net** for assistance.

Cleaning Up Conda Cache and Unused Packages

Use the `conda clean` command to clean up your Anaconda directory and free up disk space. Here are some useful options:

1. Clean Up Conda Cache and Unused Packages.

```
conda clean --all
```

This removes:

- **Unused packages** from the package cache.
- **Index cache and lock files.**
- **Temporary files** that were not deleted earlier.
- **Tarballs (downloaded package files).**

2. Remove Only Unused Packages.

```
conda clean --packages
```

This deletes packages that are not linked to any environment.

3. Remove Cached Tarballs.

```
conda clean --tarballs
```

These clears downloaded package files but keeps installed packages.

4. Remove Index Cache.

```
conda clean --index-cache
```

This clears the index cache, which stores metadata about available packages.

5. Check What Will Be Deleted Before Running.

```
conda clean --dry-run --all
```

This shows what would be removed without actually deleting anything.

For more details, visit:

<https://docs.conda.io/projects/conda/en/latest/commands/clean.html>

Stack Overflow discussion:

<https://stackoverflow.com/questions/48706548/how-to-free-disk-space-taken-up-by-anaconda>

Logging in to Intel vLab and enter lava environment in VM (lava_env) (For Windows)

1. Can be done in Windows Powershell or Anaconda. Enter these terminals and type the following.

```
ssh ncl-edu.research.intel-research.net
```

- From there `<vLab username>@<vm name>:~$` should show up.

2. After performing Step 1, access the virtual environment if created using the following.

```
source lava_env/bin/activate
```

3. To logout, type the following.

```
exit
```

OR

```
logout
```

To use Intel vLab with Jupyter Notebook (See this link: [Get Started with Intel Loihi 2 - Technical Support - Confluence](#) and more specific steps shown below.)

1. Pick a number between 10000 and 15000. You will use this number for your Jupyter notebook port. If you encounter an error indicating that the port is already in use, return to this step and pick a new number (Use port 10001.)

2. Connect to your **vLab** VM with port forwarding as follows:

```
ssh ncl-edu.research.intel-research.net -L 10001:localhost:10001
```

3. Activate virtual environment.

```
source lava_env/bin/activate
```

To Deactivate:

```
deactivate
```

4. Install Jupyter: pip install jupyterlab or pip install notebook

5. Run a Jupyter notebook or lab server:

```
jupyter notebook --no-browser --port=10001
```

OR

```
jupyter lab --no-browser --port=10001
```

~~6. Make sure to set up the environment systems to use Loihi 2 with SLURM inside vLab.~~

```
# SLURM should always be set to 1 on vLab
export SLURM=1

# LOIHI_GEN can be N3B3 or N3C1 depending on available systems
export LOIHI_GEN=N3B3

# PARTITION should be oheogulch

# Try command sinfo to see available partitions
export PARTITION=oheogulch
```

Note: As of 1/1/2026, Intel INRC Loihi Cloud service infrastructure has been taken down. Loihi 2 Hardware Tests can be performed with a Physical Hardware Chip. Intel INRC can still be used for Loihi 1 and Loihi 2 simulation tests. Please reach out to info@axialcomputing.ai for information on other means to access Loihi research systems.

7. Open a browser on your local machine and copy the following.

```
http://localhost:10001/tree?token=<token value>
```

or

```
http://127.0.0.1:10001/tree?token=<token value>
```

How to copy files/folders from local machine to Intel vLab VM.

1. Check if a directory destination in Intel VM exists by running the following.

```
ls -ld /home/<directory destination>
```

2. Use the `scp` (secure copy) and `-O` command from your terminal or Anaconda Prompt.

For individual file(s) (If wanting to place directly to home directory, copy the commands below but replace /home/<directory destination> with ~/. If using a path from Windows, copy that and surround it with quotation marks.

```
scp -O /path/to/local/file.txt <vLab username>@<vm-name>.research.intel-research.net:/home/<directory destination>
```

For Folder(s) (Use the `-r` flag.)

```
scp -O -r /path/to/local/folder <vLab username>@<vm-name>.research.intel-research.net:/home/<directory destination>
```

How to copy files/folders from Intel vLab VM back to local machine.

1. Exit jupyter notebook if not completed yet and go back to Intel vLab VM.
2. In the terminal cell or notebook, enter the following shown below. This will create the zip file for that folder.

```
cd ~  
  
# Replace folder_name with the actual folder.  
  
zip -r (folder_name).zip (folder_name)/
```

3. (Two Options)
 - a. Reopen Jupyter Notebook and (folder_name).zip will be found. Click on the box and download from the browser
 - b. Stay on Intel VM and use SCP to transfer it locally.

```
scp username@remote-host:/homes/username/(folder_name).zip  
~/Downloads/
```

Installing lava_proc

1. Obtain personal access token from Git.
(Save Git Access Token in a safe location)
2. Generate SSH key from the following

```
ssh-keygen -t ed25519 -C <your_email> research group>
```

Replace <your_email> with the email used

3. Title: Intel vLab lavaproc

Key: `ssh-<insert-key-here> <your_email>`

Select Authentication Key

NxKernel and NxCore:

To get to NXcore and NxKernel tutorials

```
/nfs/ncl/releases => Jupyter Notebook => Intel nxcore==2.5.14  
and Intel nxkernel==0.3.0
```

To activate nxkernel_env:

```
source ~/venvs/nxkernel_env/bin/activate
```

Setting up environment variables:

- Set LOIHI_GEN in Your Shell

```
export LOIHI_GEN=N3B1 # or N3B2, N3B3, N3C1, N3D1
```

- (Tip): Check which one matches your hardware or simulation setup by inspecting the documentation or asking your INRC admin.
- (Optional) To avoid setting this up every time, do the following.

```
echo 'export LOIHI_GEN=N3B1' >> ~/.bashrc
source ~/.bashrc
```

- Verify It's Set by running

```
echo $LOIHI_GEN
```

- Should see variable setup (N3B1, N3B2, N3B3, N3C1, or N3D1)
- Then re-run notebook or script, and N3Board() should initialize

If I want to set up my Intel Loihi simulation on another laptop how do I do this? On another windows operating system (upgrade to Windows 11)?

It is entirely possible to set up **Loihi simulation environment** on a *different laptop*.

Two Methods:

1. **If you want to run Loihi *simulation* locally** (Lava CPU backend)
2. **If you want to run Loihi *hardware* remotely** (physical Loihi 2 board)

Both work perfectly on Windows 11.

1. Running Loihi *Simulation* on a New Windows 11 Laptop (Check first. May NOT need to do all steps.)

Step 1 — Install Python (3.8–3.11) (From python.org; Don't do this if it's been transferred into Windows 11)

```
python --version
```

Step 2 — Install pip (if not included)

Most Python installers include pip automatically. Check:

```
pip --version
```

If missing:

```
python -m ensurepip --default-pip
```

Step 3 — Install Lava (CPU backend)

This gives you the full Loihi simulation environment.

```
pip install lava-nc
```

Installs: Lava core, CPU simulation backend, Process models (LIF, Dense, RingBuffer, etc.)

- Tools for debugging and visualization

Step 4 — Test your installation by running

```
import lava
print(lava.__version__)
```

Then run a simple LIF neuron example from Lava docs.

Step 5 — Install Jupyter Notebook

```
pip install notebook
jupyter notebook
```

2. Running Loihi 2 *Hardware* from a New Laptop

You connect to a **host machine** that controls the Loihi 2 board.

Step 1 — Generate a new SSH keypair

```
ssh-keygen -t rsa -b 4096 -f %USERPROFILE%\.ssh\loihi2_key
```

Step 2 — Send the public key (loihi2_key.pub) to the Loihi admin.

They will add it to the host machine.

Step 3 — Add a clean SSH config entry.

Open:

```
notepad %USERPROFILE%\.ssh\config
```

Add:

```
Host loihi2
    HostName <hostname or IP of the Loihi host>
    User <your username>
    IdentityFile ~/.ssh/loihi2_key
```

Step 4 — Connect

```
ssh loihi2
```

Once connected, you run Lava on the **host machine**, not your laptop.

3. Do you need to delete anything from your old laptop?

No. The old VLAB keys and old environment do **not** affect your new setup. You can start fresh on the new Windows 11 machine without touching the old one.

Summary

To set up Loihi simulation on a new Windows 11 laptop:

1. Install Python
2. Install pip
3. Install Lava (pip install lava-nc)
4. Install Jupyter if needed
5. Generate new SSH keys for hardware access
6. Add a clean SSH config entry

No need for VLAB keys, and no need to uninstall anything from the old laptop.

If I upgrade to Windows 11 on this laptop, will my Intel Loihi simulation and API keys be saved or do I have to get new API keys (more precicely the ssh ncl-edu.research.intel-research.net -L 10001:localhost:10001 session survive)?

Upgrading to Windows 11 will NOT delete your Loihi simulation environment or your API/SSH keys — unless you wipe the disk.

Breakdown of what survives and what doesn't.

Survives:

1. A standard in-place upgrade keeps **all your files**, including:
 - SSH key home directory
 - `C:\Users\<yourname>\.ssh\`
 - Windows keeps this folder during an upgrade.
 - So VLAB keys, INRC keys (if you had them), and any Loihi keys **remain intact**.
2. Python installation (Python, pip, Lava, Jupyter)

3. Loihi simulation code: Everything in Documents, Downloads, Projects stays exactly where it is.
4. SSH config is preserved `C:\Users\<yourname>\.ssh\config`
5. Environment variables: PATH additions are kept (Python, pip, etc.)

Does Not Survive: This will only happen if any of Environmental variables, SSH Configuration, Loihi Simulation Code, Python Installation is manually removed.

If you do a normal Windows 11 upgrade: You keep Loihi simulation, Python, Lava, Jupyter, SSH keys, API keys, and code

Other Links:

[Intel “Reverse Engineers the Brain” With Loihi 2 Neuromorphic Chip and Lava Framework - News](#)

<https://lava-nc.org/>

<https://github.com/lava-nc/lava>

<https://open-neuromorphic.org/neuromorphic-computing/software/snn-frameworks/lava/>